

Karim El-Sayed

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EDUCATION

Imperial College London | *MSc Computing (Distributed Systems)* 2018–2019

- Graduated with Distinction (78% average); thesis on "Optimizing React concurrent rendering for low-latency financial dashboards" published at PeerJ Computer Science, achieving 340+ citations
- Led the Imperial Coding Society as Secretary (2018–2019), grew membership from 180 to 480+ members and organized 8 industry speaker events including talks by engineers from Google Brain and DeepMind

University of Amsterdam | *BSc Computer Science* 2014–2017

- Graduated top 8% of cohort (GPA 3.87/4.0); honors thesis on "Node.js cluster optimization for microservices" selected for publication in IEEE Transactions on Software Engineering
- President of the Open Source Community Club (2016–2017), mentored 12 junior developers and maintained 3 active packages with 50K+ combined monthly downloads on npm

EXPERIENCE

Stripe | *Senior Full-Stack Engineer* 2021–Present

- Architected and shipped a new payments reconciliation dashboard in React + Node.js (GraphQL), serving 40K+ daily active merchants; reduced transaction settlement latency by 340ms (42% improvement) through optimized database queries and Redis caching, saving \$2.1M annually in payment processing delays
- Led migration of 12 legacy REST endpoints to GraphQL, reducing frontend API payload sizes by 64% (avg. 340KB → 120KB per request) and decreasing Time To Interactive (TTI) from 2.8s to 1.1s across 15M+ monthly users
- Mentored 6 junior engineers across 3 continents; designed and taught a 6-week full-stack Node.js masterclass, improving team throughput by 28% and reducing code review cycles from 4 days to 1.2 days

Figma | *Full-Stack Engineer II* 2019–2021

- Built real-time collaboration engine for Figma's design files using WebSockets + Node.js, supporting 8K+ concurrent users per server with <50ms latency; improved multiplayer sync accuracy from 94.2% to 99.7% by implementing CRDT-based conflict resolution
- Optimized React component rendering pipeline using React.memo, useMemo, and code-splitting; reduced bundle size from 780KB to 240KB (69% reduction), achieving 3.2s initial load time (Lighthouse Performance: 94), directly attributing to 12% increase in daily signups
- Owned infrastructure for multi-region deployment (AWS EC2, Kubernetes); implemented load balancing and auto-scaling that reduced p99 response time from 890ms to 180ms under peak load (8000 req/s), and achieved 99.98% uptime SLA

Shopify | *Full-Stack Engineer* 2017–2019

- Developed customer analytics dashboard (React frontend, Node.js + Express backend) used by 120K+ store owners; engineered data pipeline processing 2.1B+ events/day, cut query response times from 8.2s to 340ms through indexing and query optimization
- Contributed to Polaris design system; created 8 reusable React components used across 50+ internal applications, reducing UI development time by 35% and standardizing accessibility (WCAG 2.1 AA compliance) across Shopify's platform
- Built and maintained Node.js webhook delivery system handling 580M+ webhooks/month with 99.97% delivery success rate; designed async queue architecture using Bull.js that scaled from 2K to 45K throughput (webhooks/sec) without service degradation

Wise | *Junior Full-Stack Engineer* 2015–2017

- Implemented real-time currency exchange rate feeds using Node.js EventEmitter and WebSocket, delivering sub-500ms updates to 200K+ concurrent users; reduced customer churn by 8% through improved UX responsiveness
- Created React-based admin dashboard for monitoring transfers across 180+ currency corridors; engineered backend REST API (Node.js + PostgreSQL) handling 12K req/s with <200ms latency at p99
- Optimized database queries across 15 high-traffic endpoints, reducing median query time from 920ms to 120ms (87% improvement); implemented connection pooling and query caching, increasing throughput from 3.2K to 18K req/s

PROJECTS

LLMCache | *React, Node.js, Redis, TypeScript* 2023–Present

- Open-source semantic caching layer for LLM API calls (Node.js + Redis backend); reduces redundant API calls by 67% through vector similarity matching, cutting inference costs by \$340K/month for enterprise users
- GitHub: 4,200+ stars, 210+ forks; 18K+ weekly npm downloads; benchmarks show 5.2x faster response times vs. traditional key-value caching; featured on ProductHunt with 2.8K upvotes

ReactFlow Pro | *React, TypeScript, Canvas API* 2022–Present

- Visual node-and-edge graph editor library for React; 6,800+ GitHub stars, 340+ contributors, 92K+ weekly downloads; used by Zapier, Retool, and 15+ Fortune 500 companies for workflow automation
- Optimized rendering with virtual scrolling and WebWorker offloading; achieves 60fps on 10K+ node graphs (96% rendering improvement), and <80ms interaction latency even with complex node operations

ServerSync | *Node.js, TypeScript, GraphQL, PostgreSQL* 2021–2022

- End-to-end encrypted document sync engine; 1,200+ GitHub stars; enables offline-first applications with automatic conflict resolution and merkle-tree based delta compression
- Reduces network bandwidth by 78% through binary delta encoding; syncs 50MB+ documents in <2.1s over 3G networks; used in production by 3 privacy-focused startups serving 180K+ users

BenchMark | *Node.js, Express, React, InfluxDB* 2020–2021

- Cloud-native performance benchmarking framework for Node.js/React applications; 820+ GitHub stars, 4.2K weekly npm downloads; automated regression detection and historical trend analysis across 12 metrics
- Integrated with CI/CD (GitHub Actions, CircleCI); tracks latency, throughput, memory usage, bundle size with <1% overhead; helped clients identify performance regressions within 5 minutes (vs. manual 2-hour investigation)

EXTRACURRICULAR

React Amsterdam & Node.js Conf EU | *Keynote Speaker & Workshop Facilitator* 2021–2023

- Delivered 2 keynote talks: "Scaling React at 100M+ Users" (React Amsterdam 2023, 1,200+ attendees, 340K+ YouTube views) and "Advanced Node.js Patterns for Distributed Systems" (Node.js Conf 2022, 680+ live attendees)
- Hosted 6 hands-on workshops teaching full-stack React + Node.js architecture; 280+ attendees across events with 4.8/5 average satisfaction rating; 1 workshop published as O'Reilly training course (12K+ enrollments)

Node.js Foundation & npm Maintainer Community | *Open Source Maintainer & Mentor* 2019–Present

- Active maintainer of 4 npm packages with 180K+ combined weekly downloads (published 28 releases across projects); processed 340+ pull requests and triaged 580+ GitHub issues, maintaining <48-hour response SLA
- Mentored 8 junior developers becoming open-source contributors; 3 mentees now maintain their own packages with 50K+ downloads; designed mentorship program adopted by Node.js Foundation for 40+ participants

TECHNICAL SKILLS

Languages – JavaScript, TypeScript, Python, SQL, GraphQL, HTML5, CSS3

Technologies – React, Next.js, Node.js, Express, GraphQL, WebSockets, PostgreSQL, Redis, MongoDB, AWS (EC2, RDS, S3, Lambda), Kubernetes, Docker, Git

Tools – VS Code, Chrome DevTools, Jest, Webpack, Vite, GitHub Actions, CircleCI, Datadog, New Relic, Figma, Postman, Jira, Slack